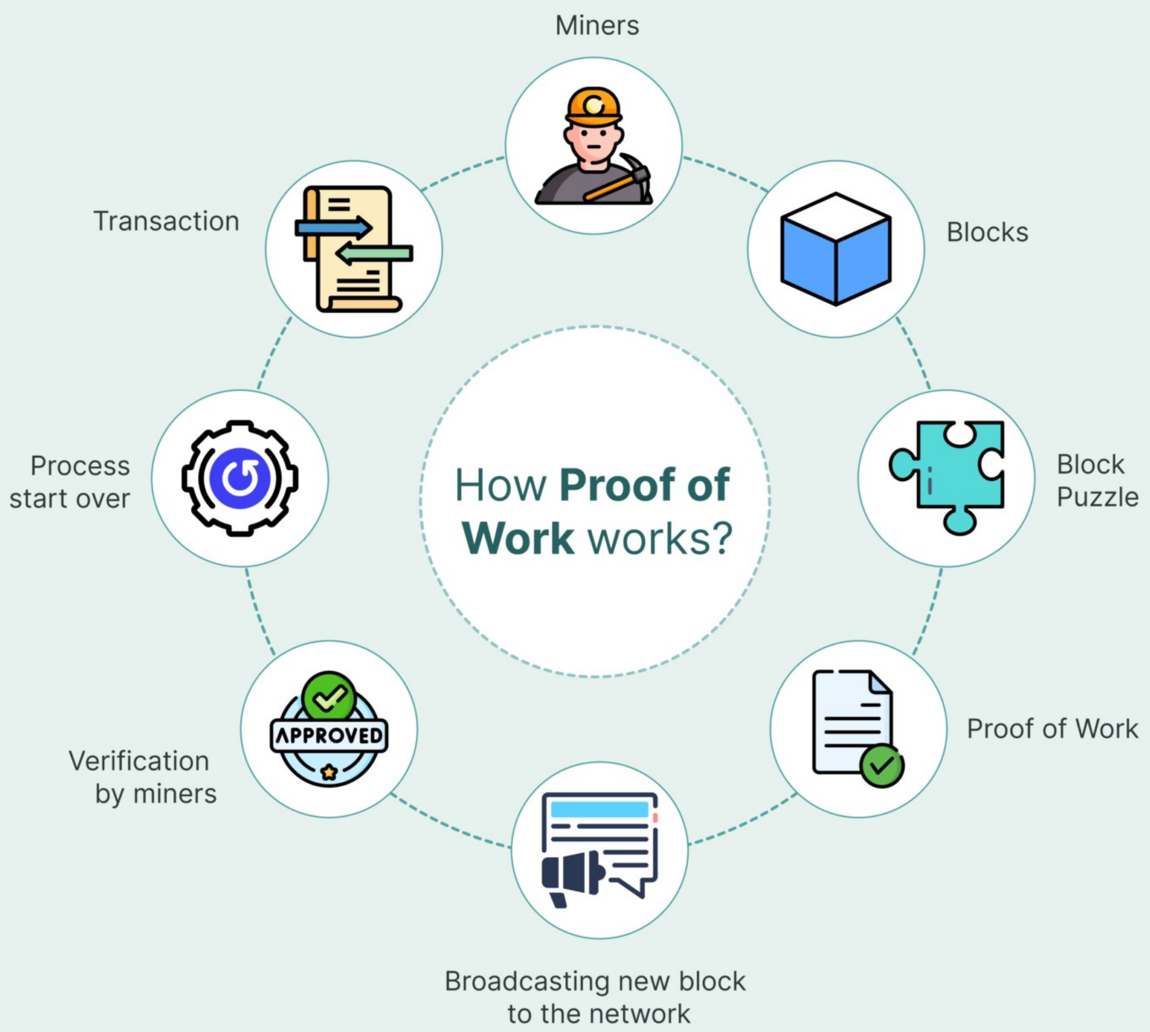
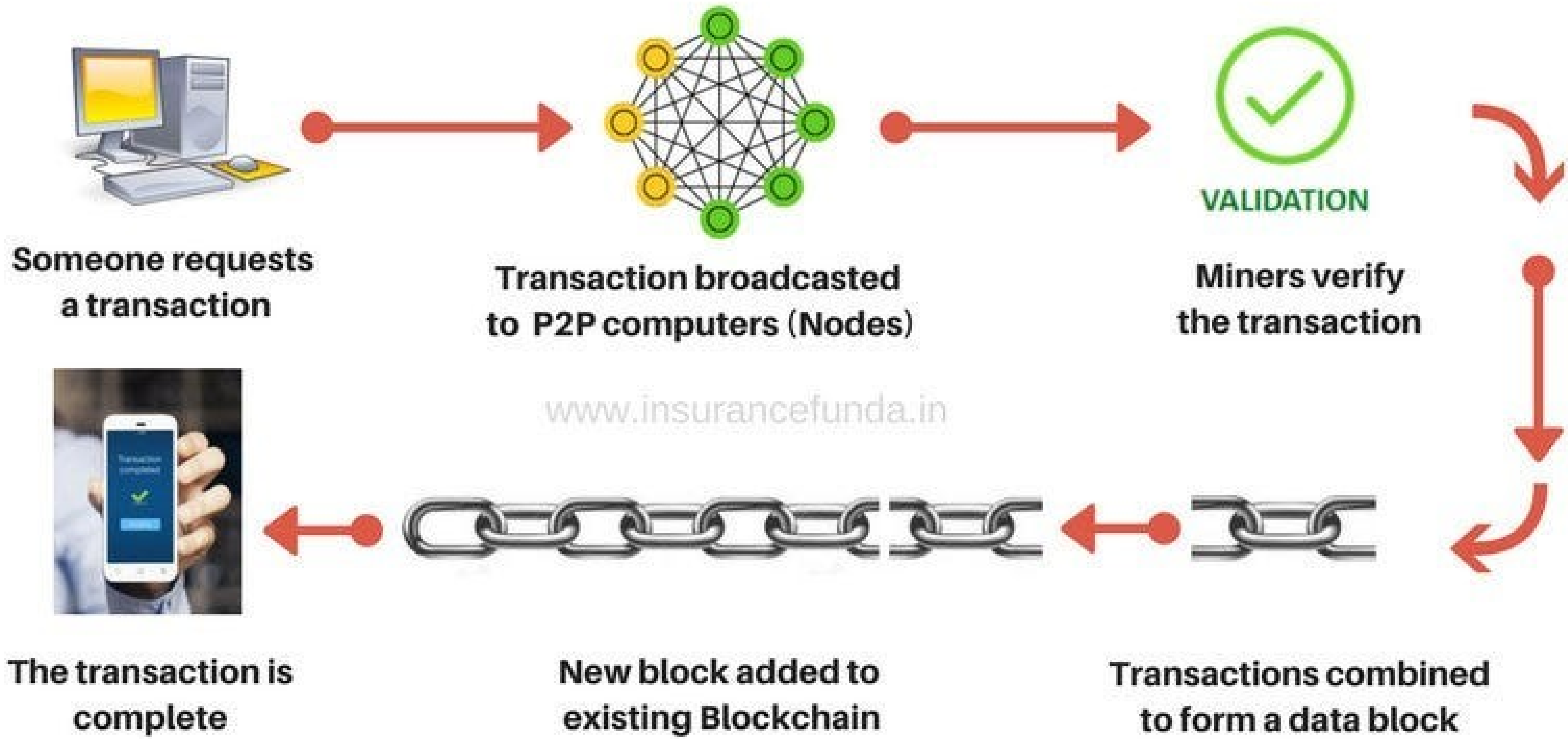
Chapter 1

Additional information for Blockchain


How Proof of Work works?
Miners
Blocks
Block Puzzle
Proof of Work
Broadcasting new block to the network
APPROVED
Verification by miners
Process start over
Transaction

HOW BITCOIN TRANSACTION WORKS



Process of proposing a New Block

1. Transaction Collection

Unconfirmed Transactions: Miners gather unconfirmed transactions from the mempool (memory pool) – a holding area for transactions that have not yet been included in a block.

2. Transaction Verification: Before mining, miners check that the transactions are valid

Validation: Digital signatures are correct., The sender has enough balance to perform the transaction.

3. Mining the Block

Hashing the Block Header: Miners start trying to find a valid hash for the block header.

The block header is passed through a cryptographic hashing algorithm (typically SHA-256 in Bitcoin).

Process of proposing a New Block

4. Validating and Broadcasting the Block

The valid hash is a proof of work, indicating that the miner has expended computational resources to find it.

5. Block Addition to the Blockchain

If the block is **valid**, it is added to the blockchain, and all nodes update their ledger.

Consensus: If The network reaches consensus on the new block, and the blockchain is updated with the latest block.

6. Reward for the Miner

The miner who successfully mined the block is rewarded with:

Block reward: Newly minted cryptocurrency (e.g., 6.25 BTC in Bitcoin as of 2024)

Mining

- Mining is the process of verifying **bitcoin transactions** and storing them in a blockchain(ledger).
- It is a process similar to gold mining but instead, it is a complicated computer code to generate a secure cryptographic system.
- The bitcoin miner is the person who solves mathematical puzzles(also called proof of work) to validate the transaction.
- The miners need **to find the perfect solution hash** that matches and fits.
- Anyone with mining hardware and computing power can take part in this.
- Numerous miners take part simultaneously to solve the complex mathematical puzzle

Who can propose a Block?

- 1. Join a Mining Pool (Most Common Method)
  - Instead of mining solo, you join a mining pool (e.g., F2Pool, Slush Pool, Antpool).
  - The pool collectively mines blocks and shares rewards among members.
  - The pool operator is responsible for proposing blocks on behalf of the miners.
- 2. Cloud Mining Services
  - You rent mining power from a company like Genesis Mining, NiceHash, or Bitdeer.
  - The company operates the mining hardware and proposes blocks.
  - You earn a share of the mining rewards based on your contract.
- 3. Mining as a Service (MaaS)
  - Some specialized firms offer Mining as a Service, where they mine on your behalf and share profits with you.
  - Similar to cloud mining but with more transparency and customizability.

Mining

- Miners are using:
  - **Specialized mining hardware: “application-specific integrated circuits(ASICs).**
  - **A Bitcoin mining software to join the Blockchain network.**
  - **Powerful GPU (graphics processing unit).**
  - **Bitcoin Wallet: To be kept the bitcoin, after the user receives bitcoins as a reward for mining**
  - **The winning miner gets a block reward (e.g., 3.125 BTC in Bitcoin)**
  - **You cannot mine PoS coins (Ethereum 2.0, Cardano) because they use staking instead of mining**

Points to Remember

Acronym	Full Form	Meaning
PoW	Proof of Work	Miners solve cryptographic puzzles to validate blocks (energy intensive).
PoS	Proof of Stake	Validators create blocks based on staked cryptocurrency.
DPoS	Delegated Proof of Stake	Token holders vote for delegates to validate transactions.
LPoS	Leased Proof of Stake	Users lease stake to validators to earn rewards.
BFT	Byzantine Fault Tolerance	Ability of a system to function despite malicious nodes.
PBFT	Practical Byzantine Fault Tolerance	Consensus mechanism for permissioned blockchains.

Threats & Challenges

- **51% Attack:** every framework has a design flaw
- The transaction is confirmed and added to BCT only if a minimum percentage of all nodes accept it
- For bitcoin originally the % was 51%
- In a public BCT **if a miner has 51% hashrate,** then they can manipulate the ledger
- **Ripple uses 80% consensus model.**

Challenges to Proof-of-Work

- **51% Attack**
- A 51% attack occurs when a malicious actor controls more than 50% of the network's mining power.
- This would allow them to manipulate transactions, double spend, or block valid transactions.
- **Orphaned Blocks**
- If a miner mines an invalid block or a block that is not chosen by the majority
- The block is considered orphaned and the miner loses the block reward.

Threats & Challenges

- **Double Spending:** One can play with cryptocurrencies trying to spend the same money twice in quick succession.
- While the transaction is evaluated by miners, and takes some time to get confirmed.
- **But before confirmation**, person may send the same amount again to another one.
- So there are 2 unconfirmed transactions in pool called double spending
- In order to avoid this:
  - BCT keeps a timestamp of each transaction

Threats & Challenges

- **Denial of Service Attack:** is a malicious attack on BlockChain by **people who send fack blocks huge in length** to BCT n/w.
- It takes **enormous amount of time for miners** to evaluate the block that reduces the transaction rate of adding blocks to the n/w.
- It happens due to:
  - BCT is public and anyone can access
  - **There is no maximum limit to size of the blocks**
- Some have created a Private / Permissioned BCT where only authorized can access the BCT

Threats & Challenges

- Currently, Miners take
  - Bitcoin (Proof of Work): almost **10 minutes** to mine a bitcoin block which is much more.
  - Ethereum (Proof of Work/Transitioning to Proof of Stake): 13-15 seconds
  - Litecoin (Proof of Work): a new block is mined every 2.5 minutes
  - Cardano (Proof of Stake): 20 seconds
  - Solana (Proof of History): 400 millisecond

Segwit

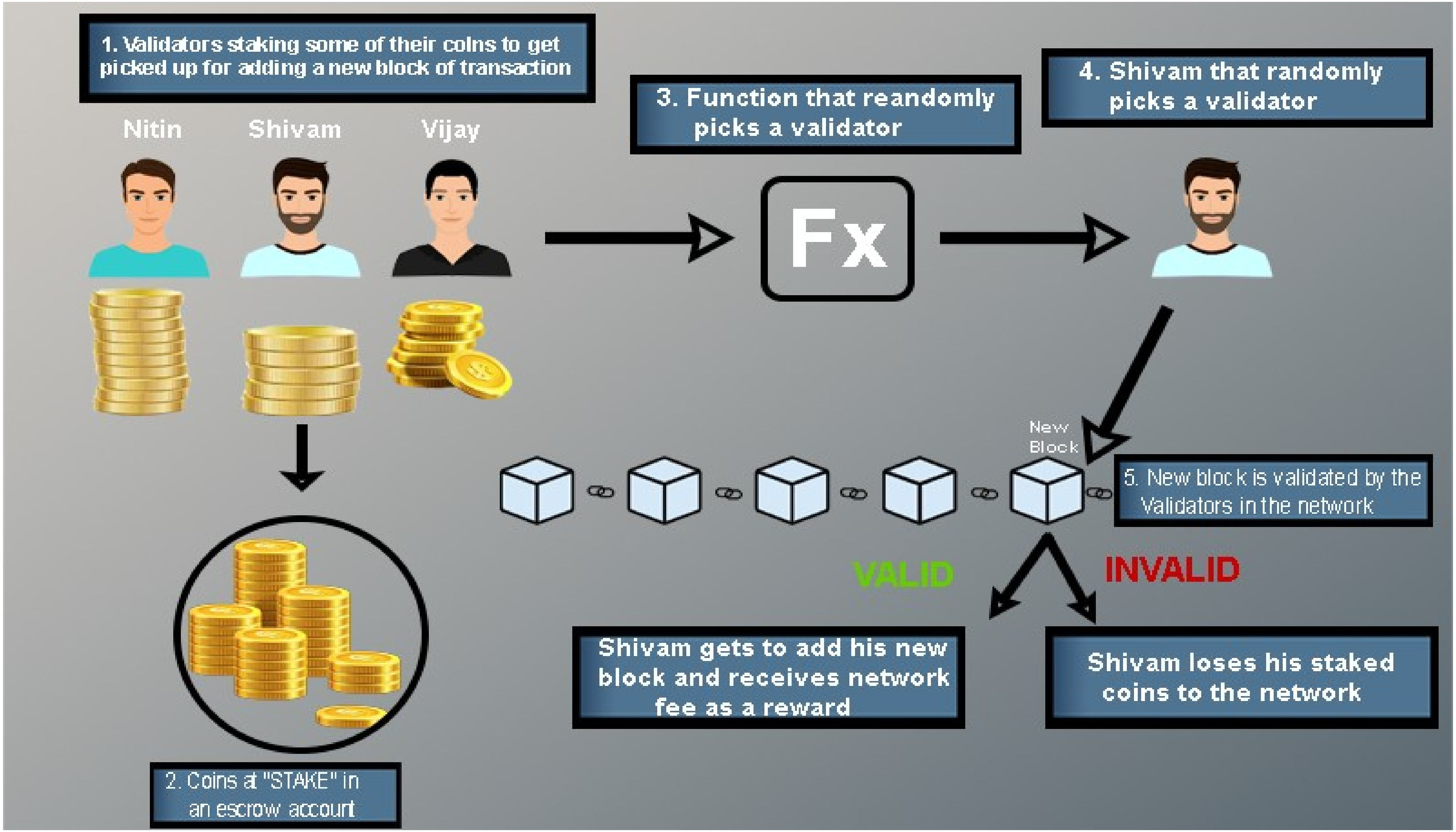
- Some Blockchainers applied a new technique called segwit (segregated Witness).
- **Solution bitcoin proposed** : Segwit **impose a restriction on block size** limiting in to only **1 MB**.
- <u>Signature part of the block called "**witness**" would move to trailing part of the block.</u>
- It makes mining much easier and faster due to small size of blocks
- This solution also implemented in crypto-currencies like:
  - Groestlcoin, Litecoin, DigiByte and Vertcoin
- **Segwit2X**: In november 2017, Segwit2X activated:
  - to increase the block size to 2 MB

Proof-of-Work

- Original consensus algorithm(mathematical Puzzle)
- **Miners compete against each other** to complete transactions on the network
- The right solution **they announces** it to the whole n/w
- Receiving a crypto currency prize provided by the protocol PoW.
- Miners use more and more powerful machines due to increase of complexity
- e.g: Bitcoin (BTC), Litecoin (LTC), Dogecoin (DOGE)

Proof-of-Stake

- Proof-of-stake (POS) was created as an alternative to proof-of-work (POW)
- While PoW mechanisms require miners to solve cryptographic puzzles,
- PoS mechanisms require validators to hold and stake tokens for the privilege of earning transaction fees.
- Under proof-of-stake (POS), validators are chosen based on the **number of staked coins they have.**
- e.g: Ethereum (ETH 2.0), Cardano (ADA), Solana (SOL).



1. Validators staking some of their coins to get picked up for adding a new block of transaction
Nitin
Shivam
Vijay
3. Function that reandomly picks a validator
Fx
4. Shivam that randomly picks a validator
New Block
5. New block is validated by the Validators in the network
VALID
INVALID
Shivam gets to add his new block and receives network fee as a reward
Shivam loses his staked coins to the network
2. Coins at "STAKE" in an escrow account

Process at Proof-of-Stake

- **Validators Locking Coins**: Users deposit a certain amount of cryptocurrency into the network.
- **Becoming a Validator**: If the stake meets the minimum requirement (e.g., 32 ETH for Ethereum 2.0), the user can validate transactions.
- **Earning Rewards**: Validators receive staking rewards for confirming transactions and securing the network.

Fees for Proof-of-Stake

- Fee Unit: Gas and Gwei

- Ethereum uses a two-level system:
  - Gas
  - Measures the computational work required.

- Example: Simple transfer ≈ 21,000 gas

- Gas Price, Paid in Gwei

- 1 ETH = 1,000,000,000 Gwei (10^9)

What are smart contracts?

- Lines of code that are stored on a blockchain
- **Automatically execute** when predetermined terms and conditions are met.
- They are programs run by the people who developed them.
- In business collaborations, used to enforce some type of agreement

The structure of any smart contract includes the following attributes:

- The contract parties that have accepted the agreed conditions (for this, an electronic signature or multi-signatures, if there are several parties, is used);
- The environment in which the contract will be located (for example, Ethereum);
- The subject of the contract – namely, resources for exchange;
- Terms of the contract – a description of the mathematically confirmed conditions under which the contract will be considered as fulfilled.

A smart contract will store all of the listed funds until the set goal is reached. If the goal is not achieved, then the money is returned to investors.

Smart contracts in Supply Chain

- Smart Contract Triggers
  - Payment release on successful delivery confirmation
  - Retailer finalizes order → Payment automatically sent to distributor.
  - If GPS tracking shows unexpected route deviation, an alert is triggered for possible theft or fraud.
  - If stock level falls below a predefined threshold, the smart contract automatically places a new order with the supplier.
  - If GPS tracking shows unexpected route deviation, an alert is triggered for possible theft or fraud.

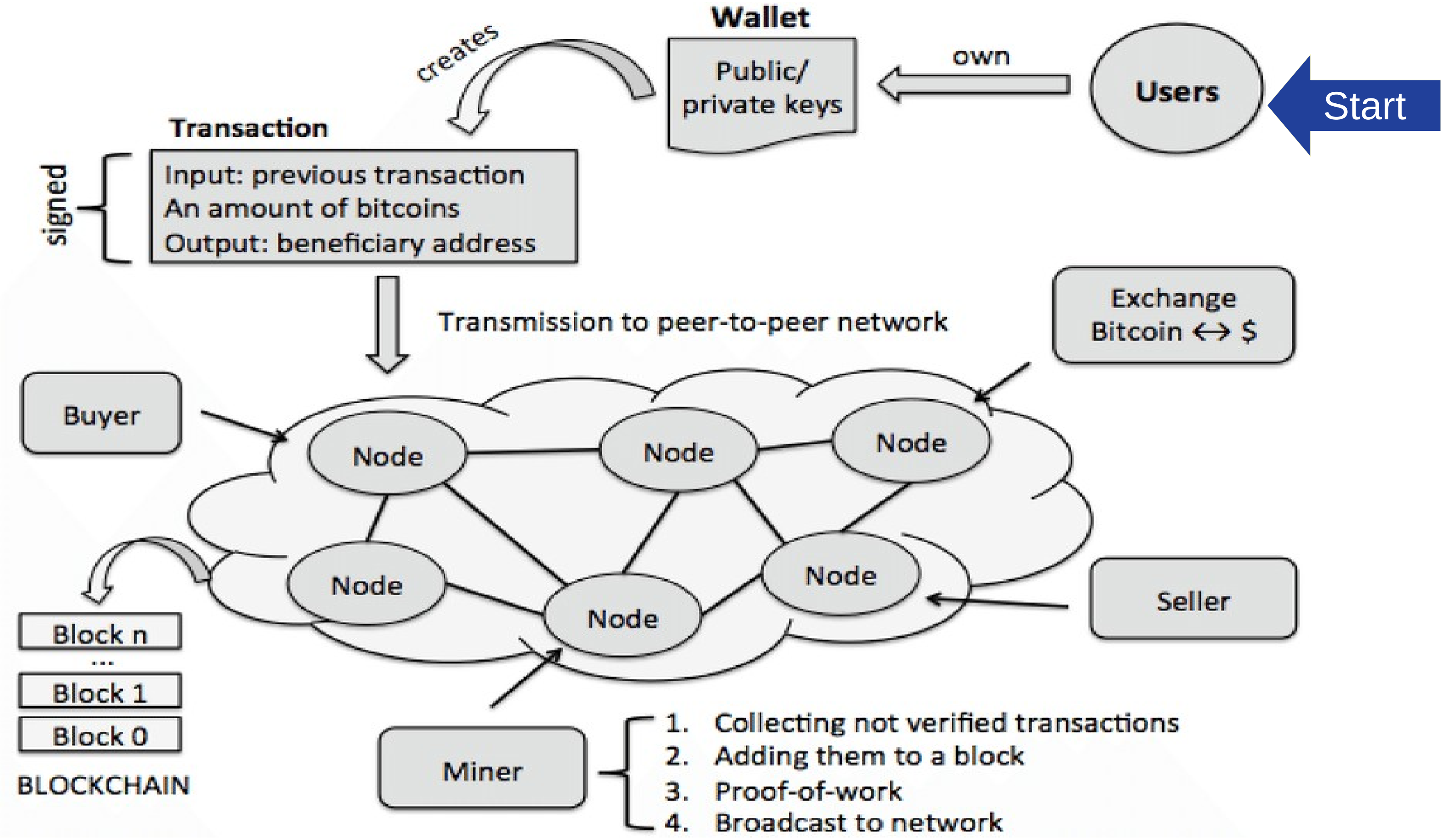
DAPPS

- A decentralized application: DApp, dApp, Dapp, or dapp
- **A computer application** that runs on a distributed computing system.
- It uses distributed ledger technologies (DLT) in various applications such as
  - DeFi - Decentralized Finance
  - **Finance:** Uniswap (Ethereum) – A decentralized exchange (DEX) that enables users to swap cryptocurrencies without intermediaries.
  - **Social Media**: Steemit(Hive blockchain) – A decentralized blogging platform where users earn rewards in cryptocurrency for content creation.
  - **Health care:** Medicalchain (Hyperledger) – A blockchain-based platform that enables secure storage and sharing of electronic health records (EHRs).
  - Many more

Other Applications of BCT

- Financial market
- Media and entertainment
- Energy trading
- Prediction markets
- Retail chains
- Loyalty rewards systems (helps businesses to meet target and give incentives to the right customers)
- Insurance
- Logistics and supply chains(transporting goods to customers)
- Medical records
- Government and military applications

Summary



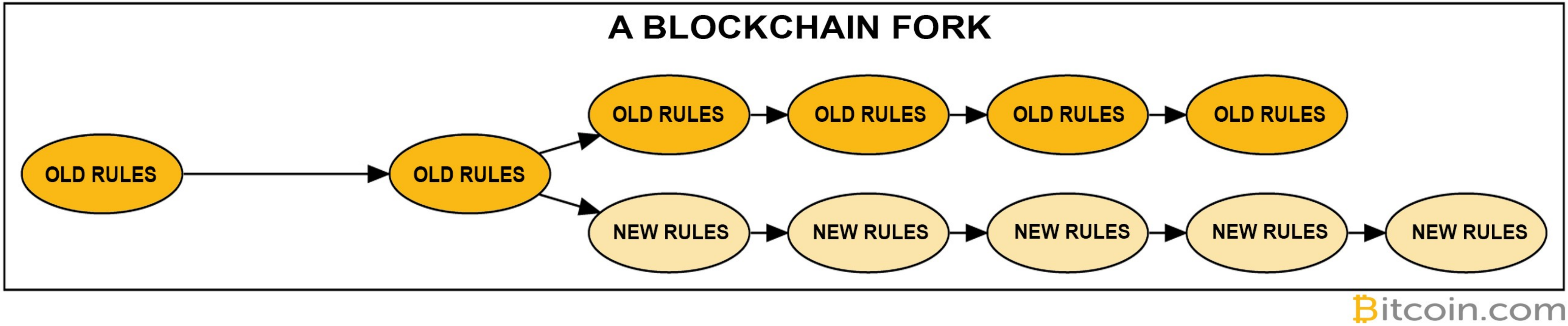
Wallet
Public/ private keys
own
Users
Start
creates
Transaction
signed
Input: previous transaction
An amount of bitcoins
Output: beneficiary address
Transmission to peer-to-peer network
Exchange Bitcoin ↔ $
Buyer
Node
Node
Node
Node
Node
Node
Seller
Block n
...
Block 1
Block 0
BLOCKCHAIN
Miner
1. Collecting not verified transactions
2. Adding them to a block
3. Proof-of-work
4. Broadcast to network


BTC
USD
579.60
574.65
0.8832
0.8828

Wallet

- A crypto-currency wallet digitally stores a <u>user’s public and private keys</u>
- Also programmatically helps in <u>sending and receiving</u> digital currency
- **Examples:**
  - **MetaMask** – A popular browser extension and mobile wallet for Ethereum and ERC-20 tokens.
  - **Trust Wallet** – A mobile wallet that supports multiple blockchains.
  - **Exodus** – A multi-asset desktop and mobile wallet with a user-friendly interface.
  - **Coinbase Wallet** – A self-custodial wallet separate from Coinbase Exchange.
  - **Electrum** – A lightweight Bitcoin wallet with strong security features.

Fork in BCT(Soft Fork & Hard Fork)



Fork in BCT(Soft Fork & Hard Fork)

- **Fork means** <u>blocks of different version of BCT</u>
- It occurs when the network's rules or protocols are updated.
- Forks can be categorized into **hard forks and soft forks**
  - based on their impact on **backward compatibility and network consensus.**
- Transactions are added to block and the new block is finally added to the BCT n/w

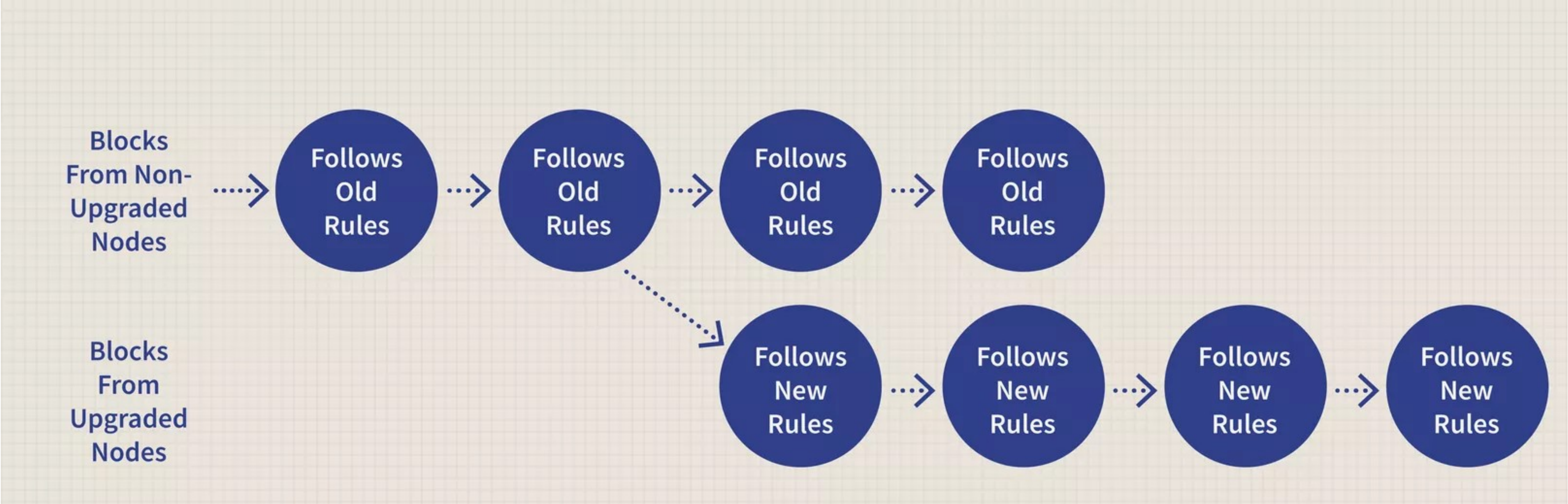
Fork in BCT(Soft Fork & Hard Fork)

- Combined effort of **miners and a distributed consensus to validate** the everyone in n/w using **same version of BCT**
- As people are using different version of BCT, In such case a temporary or accidental fork is created.
- If we use **Ethereum as BCT framework, chances are more for fork**
- However soon is sorted out by getting rid of the faulty blocks. It is called a **soft fork.**

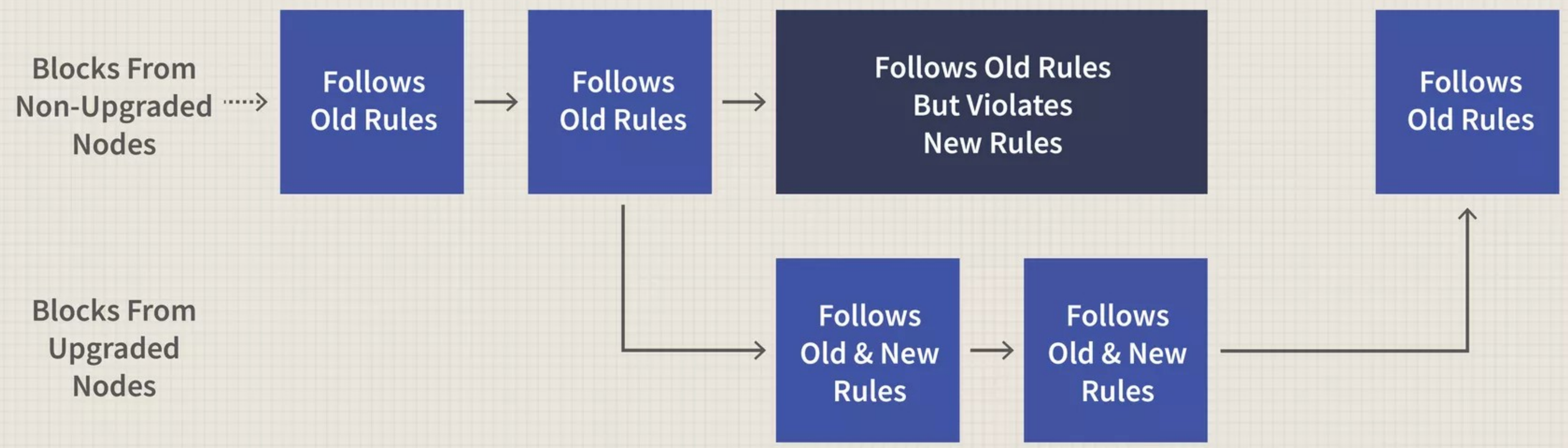
Hard Fork & Soft Fork

- From time to time there is a need in the s/w to change or upgrade
- So <u>two different versions of BCT are created</u> sharing the same origin
- It is called hard fork
- Its a Radical change, new blockchain, not backward-compatible

- A soft fork is a **change to the software protocol**
- **When old nodes will recognize the new blocks as valid**, a soft fork is backwards-compatible.
- Minor change, no new blockchain



A Hard Fork: Non-Upgraded Nodes Reject The New Rules, Diverging The Chain



A Soft Fork: blocks violating new rules are made stale by the upgraded mining majority

Feature	Hard Fork	Soft Fork
Definition	A major change that makes previous versions incompatible with the new version.	A minor upgrade that remains compatible with previous versions.
Backward Compatibility	Not backward-compatible (nodes must upgrade to continue participating).	Backward-compatible (older nodes can still validate transactions).
Consensus Requirement	Requires majority of nodes and miners to upgrade.	Can be implemented even if only a majority of miners adopt it.
Network Split	Causes a permanent split, creating two separate blockchains (old and new).	Does not cause a split, as older nodes still recognize the updated chain.
Example	Bitcoin Cash (BCH) split from Bitcoin (BTC) in 2017 due to block size increase.	SegWit (Segregated Witness) update in Bitcoin, which improved scalability without splitting the chain.
Impact on Transactions	Transactions on the new chain are not recognized by the old chain.	Transactions remain valid on both old and new nodes.
Risk Factor	Higher risk of network division and reduced security.	Lower risk, as old nodes remain functional.